

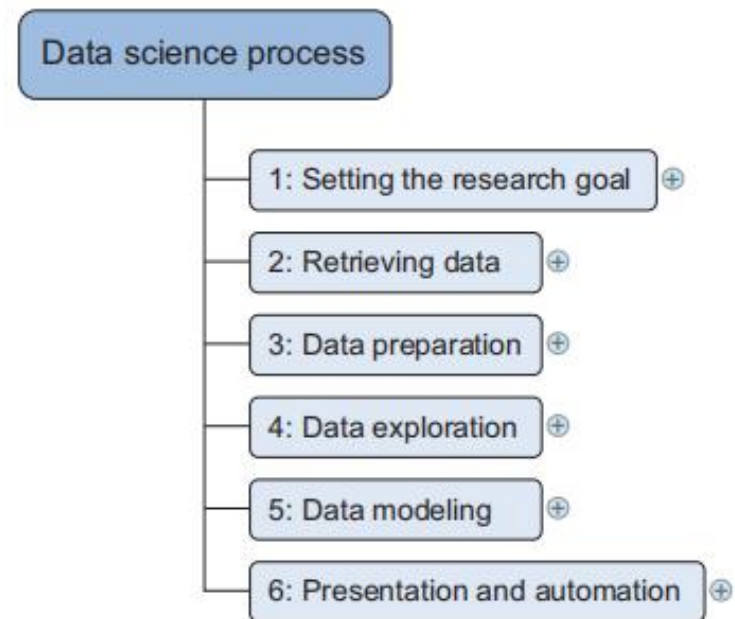
Data Science Lifecycle

Week 2

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Data Science Lifecycle

1. Problem Definition
2. Data Collection
3. Data Cleaning & Preparation
4. Exploratory Data Analysis (EDA)
5. Model Building
6. Presentation and Automation



Step 1: Setting the research goal

- Don't be a slave to the process
- Spend time understanding the goals and context of your research
- project charter requires teamwork, and your input covers at least the following:
 - A clear research goal
 - The project mission and context
 - How you're going to perform your analysis
 - What resources you expect to use
 - Proof that it's an achievable project, or proof of concepts
 - Deliverables and a measure of success
 - A timeline

Step2: Retrieving data

- Collecting, Cleaning, Analyzing and Visualizing
- Start with data stored within the company
- Don't be afraid to shop around
- Do data quality checks now to prevent problems later

Open data site	Description
Data.gov	The home of the US Government's open data
https://open-data.europa.eu/	The home of the European Commission's open data
Freebase.org	An open database that retrieves its information from sites like Wikipedia, MusicBrains, and the SEC archive
Data.worldbank.org	Open data initiative from the World Bank
Aiddata.org	Open data for international development
Open.fda.gov	Open data from the US Food and Drug Administration

Collecting Data

- Hunting
 - Companies and proprietary data sources have enormous data that can help how the world works.
 - Facebook, Google and Amazon ha
 - Public datasets
 - Kaggle: <https://www.kaggle.com/datasets>
 - UCI Machine Learning Repository: <https://archive.ics.uci.edu>
 - Data.gov (USA), Data.gov.pk (Pakistan), Eurostat, World Bank
- Scraping
 - Automated collection of data from websites
 - Spidering, process of downloading the right set of pages for analysis
 - Scraping is the fine art of stripping this content from each page.
 - Python Libraries like BeautifulSoup, Scrappy

Collecting Data

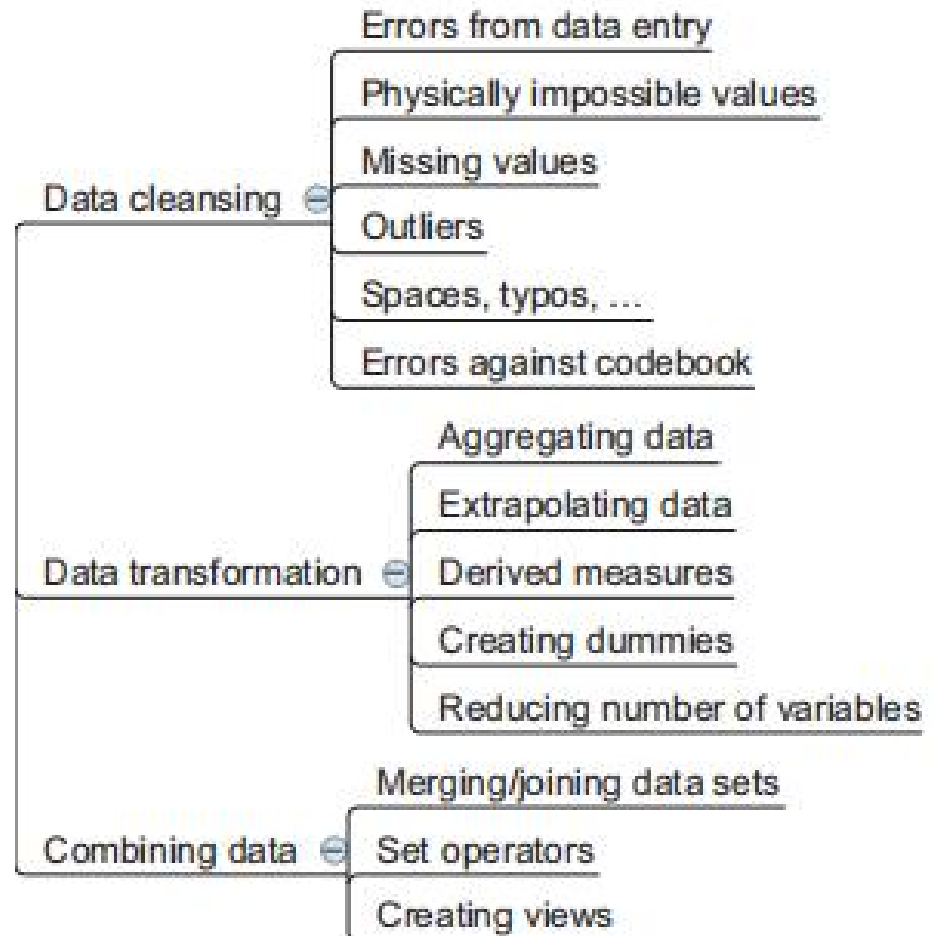
- APIs (Automated Collection)
 - Twitter API – for tweets, hashtags
 - Weather API – for temperature, humidity, forecasts
 - News API – for headlines and articles
 - OpenStreetMap API – for geographic data
- Make your own dataset
 - 3TB drive, if each records takes 1KB, the device in principle has room for 3 billion records.

Data Acquisition Methods

- Manual Collection: Surveys, forms
- Data Purchase: Paid datasets
- Streaming Data: IoT, sensor data
- Crowdsourcing platforms like Amazon Turk and CrowdFlower

Step 3: Cleansing, integrating, and transforming data

- removing errors in your data so your data becomes a true and consistent



General solution

Try to fix the problem early in the data acquisition chain or else fix it in the program.

Error description

Possible solution

Errors pointing to false values within one data set

Mistakes during data entry

Manual overrules

Redundant white space

Use string functions

Impossible values

Manual overrules

Missing values

Remove observation or value

Outliers

Validate and, if erroneous, treat as missing value
(remove or insert)

Errors pointing to inconsistencies between data sets

Deviations from a code book

Match on keys or else use manual overrules

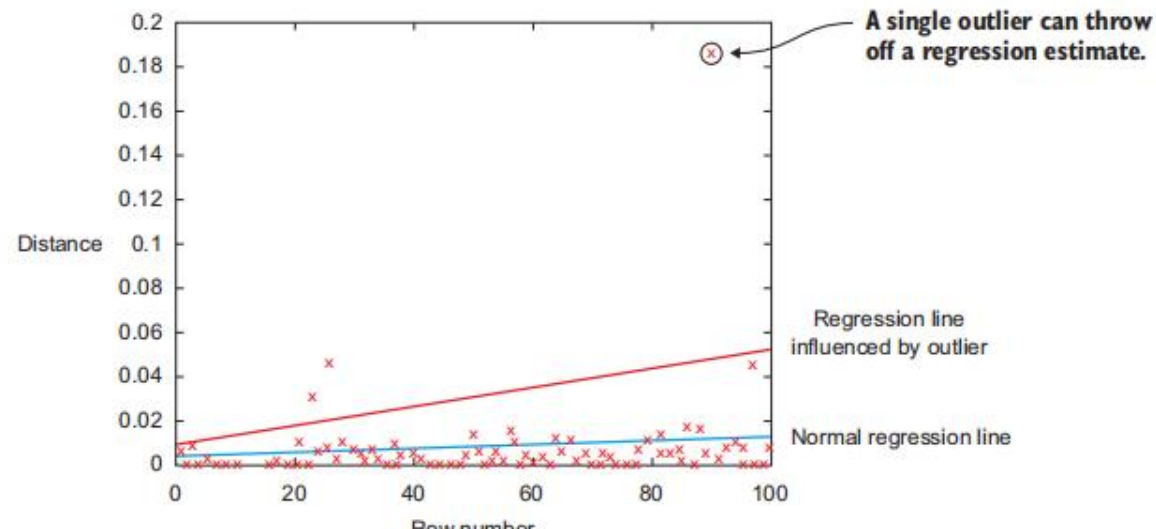
Different units of measurement

Recalculate

Different levels of aggregation

Bring to same level of measurement by aggregation
or extrapolation

- For example, in figure, we use a measure to identify data points that seem out of place. We do a regression to get acquainted with the data and detect the influence of individual observations on the regression line. When a single observation has too much influence, this can point to an error in the data, but it can also be a valid point.
- At the data cleansing stage, these advanced methods are, however, rarely applied and often regarded by certain data scientists as overkill.



Data Cleaning-

- Data collection and data entry are error-prone processes. They often require human intervention, and because humans are only human, they make typos or lose their concentration for a second and introduce an error into the chain.

Value	Count
Good	1598647
Bad	1354468
Godo	15
Bade	1

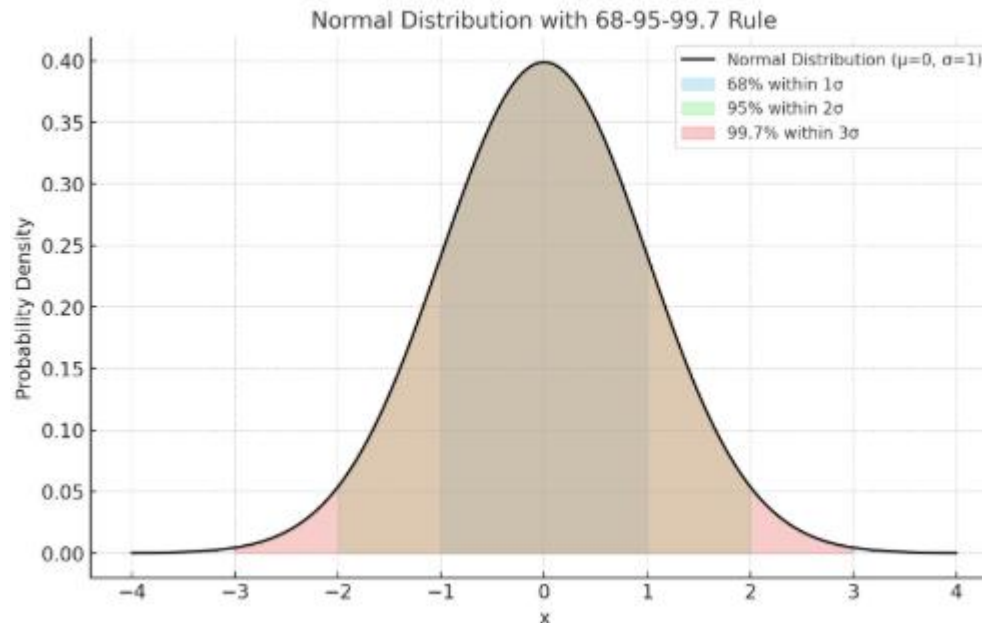
```
if x == "Godo":  
    x = "Good"  
if x == "Bade":  
    x = "Bad"
```

Data Cleaning

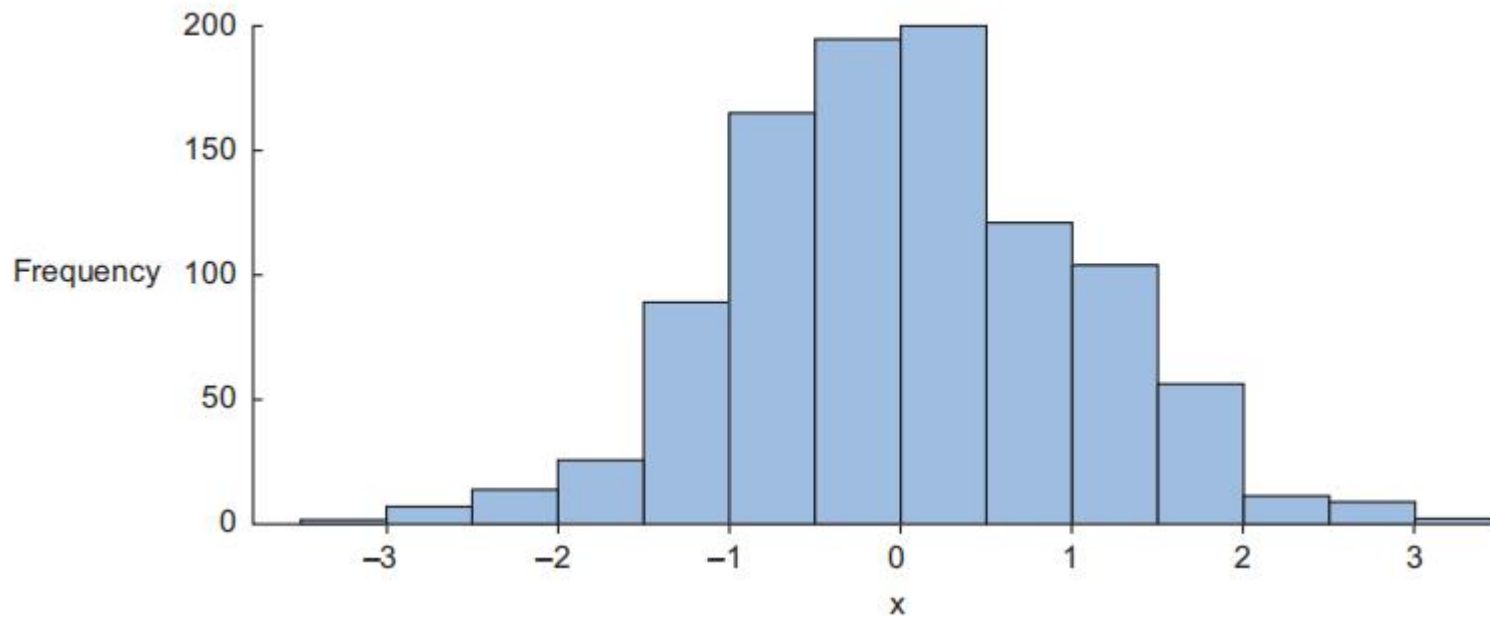
- Redundant white spaces
 - string functions that will remove the leading and trailing whitespaces.
 - Python has strip() function
- Fixing capital letters mismatches
 - by applying a function that returns both strings in lowercase, such as .lower() in Python. "Brazil".lower() == "brazil".lower() should result in true.
- Impossible values and Sanity Checks
 - against physically or theoretically impossible values such as people taller than 3 meters or someone with an age of 299 years. Sanity checks can be directly expressed with rules:
check = 0 <= age <= 120

Data Cleaning-Outliers

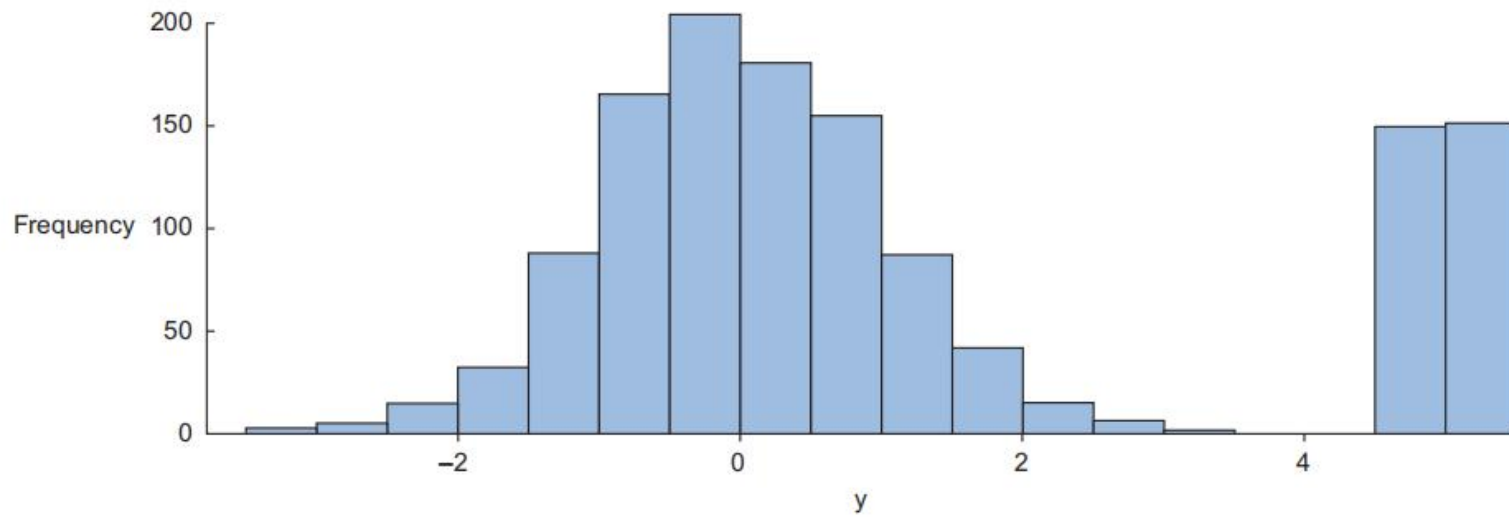
- one observation that follows a different logic or generative process than the other observations.
- The easiest way to find outliers is to use a plot or a table with the minimum and maximum values.



Expected distribution



Distribution with outliers



Data Cleaning- Dealing with missing values

Technique	Advantage	Disadvantage
Omit the values	Easy to perform	You lose the information from an observation
Set value to null	Easy to perform	Not every modeling technique and/or implementation can handle null values
Impute a static value such as 0 or the mean	Easy to perform You don't lose information from the other variables in the observation	Can lead to false estimations from a model
Impute a value from an estimated or theoretical distribution	Does not disturb the model as much	Harder to execute You make data assumptions
Modeling the value (nondependent)	Does not disturb the model too much	Can lead to too much confidence in the model Can artificially raise dependence among the variables Harder to execute You make data assumptions

Data Cleaning

- Deviations from a code book
 - A code book is a description of your data, a form of metadata. It contains things such as the number of variables per observation, the number of observations, and what each encoding within a variable means.
- Different units of measurements
 - When integrating two data sets, you have to pay attention to their respective units of measurement.
- Different levels of aggregation
 - An example of this would be a data set containing data per week versus one containing data per work week.

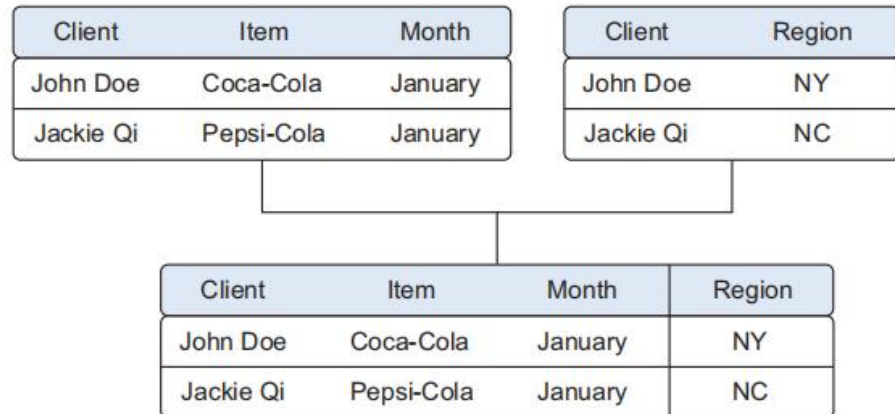
Correcting errors

- Not everyone spots the data anomalies. Decision-makers may make costly mistakes on information based on incorrect data.
- If errors are not corrected early on in the process, the cleansing will have to be done for every project that uses that data.
- Data errors may point to defective equipment, such as broken transmission lines and defective sensors.
- Data errors can point to bugs in software or in the integration of software that may be critical to the company.

Integrating data

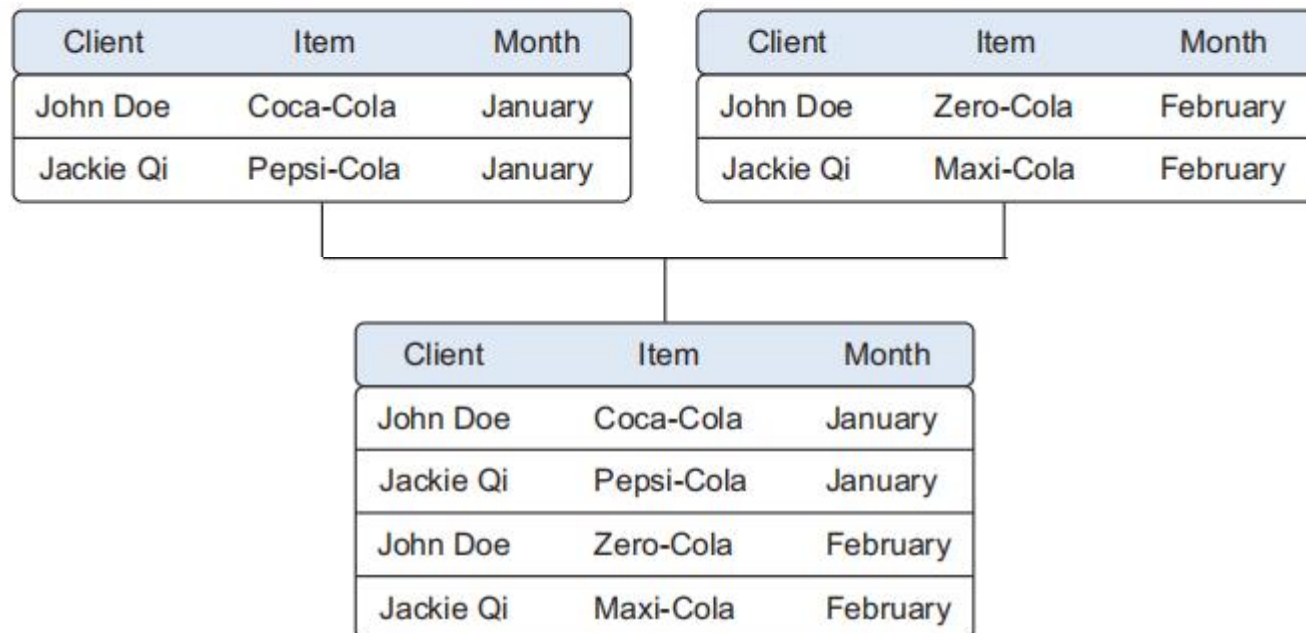
- **Joining**

- enriching an observation from one table with information from another table.



- **Appending or stacking**

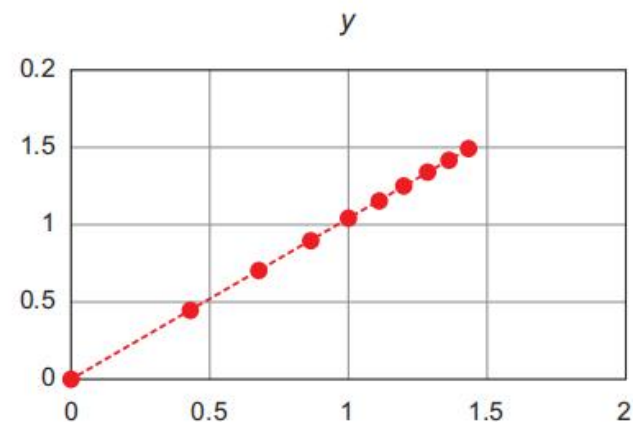
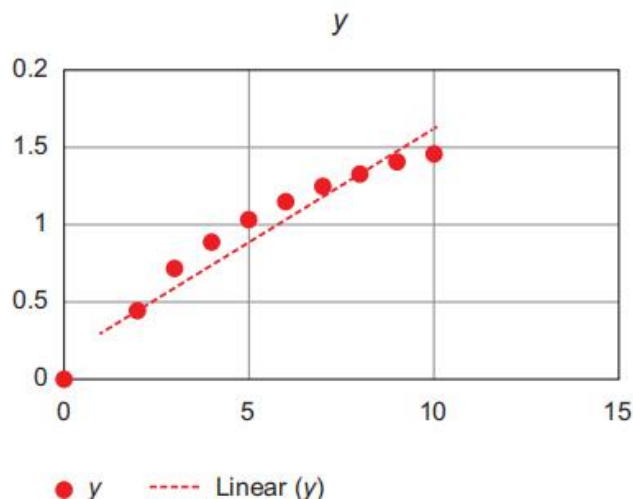
- adding the observations of one table to those of another table.



Transforming data

- Certain models require their data to be in a certain shape.
- Relationships between an input variable and an output variable aren't always linear

x	1	2	3	4	5	6	7	8	9	10
$\log(x)$	0.00	0.43	0.68	0.86	1.00	1.11	1.21	1.29	1.37	1.43
y	0.00	0.44	0.69	0.87	1.02	1.11	1.24	1.32	1.38	1.46



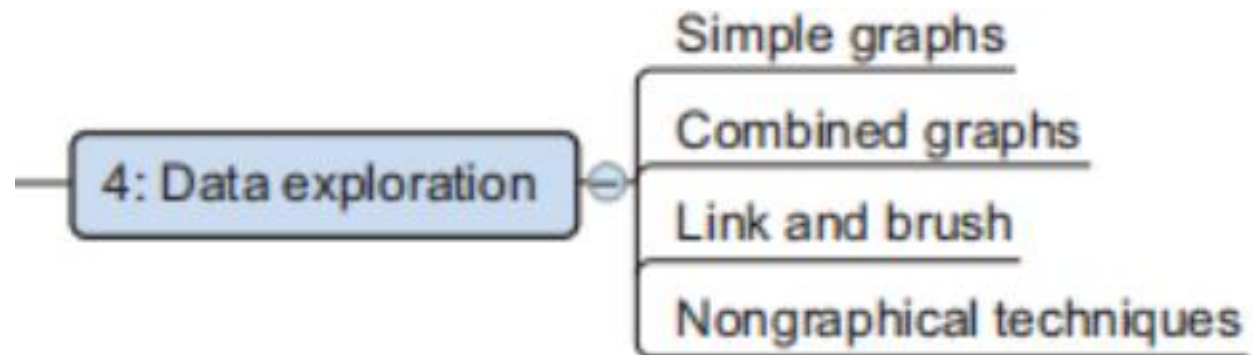
Transforming data

- Reducing the number of variables
 - Data scientists use special methods to reduce the number of variables but retain the maximum amount of data.
- Turning variable into dummies

Customer	Year	Sales	Male	Female
1	2015	10	0	1
1	2016	11	0	1
2	2015	8	1	0
3	2016	12	1	0
3	2017	13	1	0
4	2017	14	0	1

Step 4: Data Exploration (EDA)

- Information becomes much easier to grasp when shown in a picture, therefore you mainly use graphical techniques to gain an understanding of your data and the interactions between variables.
- With brushing and linking you combine and link different graphs and tables (or views) so changes in one graph are automatically transferred to the other graphs.

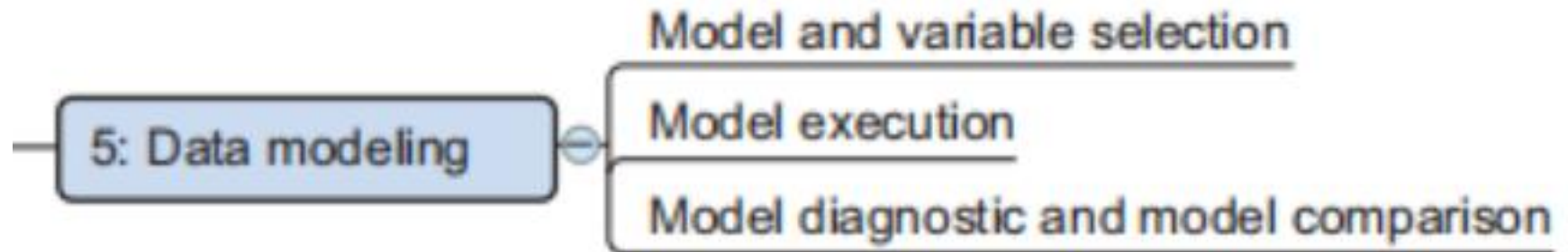


Lets Explore Pandas and Numpy

- DataFrame is a 2D labeled data structure in pandas (like a table in Excel).
- It has rows and columns.
- Each column can have a different data type (int, float, string, etc.)

Step 5: Data Modeling

- Your findings from the exploratory analysis should already give a fair idea of what variables will help you construct a good model.



Step 6: Presentation and Automation

- After you've successfully analyzed the data and built a well-performing model, you're ready to present your findings to the world .